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Final Summary Table for Unit 1 :

Unit 1: Information Storage and Management Summary Table

Topic	Key Idea	Example	Importance
Importance of Data Storage	Digital data is a company's most critical resource, representing years of gathered knowledge and intelligence.	Storing **healthcare records** (X-rays, patient histories) or processing **airline reservations**.	Loss of data can lead to catastrophic business failure; it is the central pillar for 24x7 global operations.
Business Issues and IT Challenges	IT managers face a "perfect storm" of exploding data volumes and costs while lacking enough skilled professionals.	Managing the rapid growth of **unstructured data** (videos, emails, photos), which accounts for 80% of enterprise data.	Failure to address these leads to regulatory fines (e.g., **GDPR, HIPAA**) and poor service availability.
Server and Storage I/O Fundamentals	Data flows from applications to storage via a system of hosts, connectivity (HBAs), and storage media.	An **Initiator** (server) starting an I/O request and a **Target** (disk) responding to it.	Understanding disk performance laws (e.g., **Little's Law**) prevents bottlenecks that cause application latency.
Virtualization and Storage Services	Abstraction of physical resources to make them appear as logical resources to increase utilization and flexibility.	Cloud service models like **IaaS** (raw storage/servers), **PaaS** (dev tools), and **SaaS** (hosted apps).	Enables **elasticity and multitenancy**, allowing businesses to scale resources on demand without capital investment.
Data and Storage Access	Mechanisms for interacting with data: block-level (addresses), file-level (names), or object-level (unique handles).	Using a **SAN** for block-level database access or a **NAS** for simple file sharing via NFS/CIFS.	Aligns storage architecture with specific application needs (e.g., high-speed databases vs. unstructured file sharing).
Infrastructure and Resource Management	Best practices and tools focused on four key metrics:	**Information Lifecycle Management (ILM)**, which moves data	Maximizes the return on investment and ensures that IT

	Performance, Availability, Capacity, and Energy (PACE).	between tiers based on its value over time.	infrastructure aligns with business service levels.
Data Movement and Migration	The process of moving data between systems for consolidation, technology upgrades, or disaster recovery.	**Non-disruptive migration** where virtual machines (e.g., VMotion) stay online while data is moved between physical devices.	Minimizes downtime during upgrades and allows for agile resource balancing across the data center.
I/O Connectivity and Networking Fundamentals	The physical and logical layers that connect servers to storage, including cables, ports, and protocols.	**Fibre Channel (FC)** for high-speed SANs or **iSCSI** for sending storage commands over standard Ethernet.	Provides the high-bandwidth, low-latency foundation required for reliable and scalable data movement.